SUBMITTAL

1) Outdoor Ductwork Insulation
2) Outdoor weather proof jacketing

PROJECT:

CONTRACTOR: Northern Wolves

SUBMITTED BY:

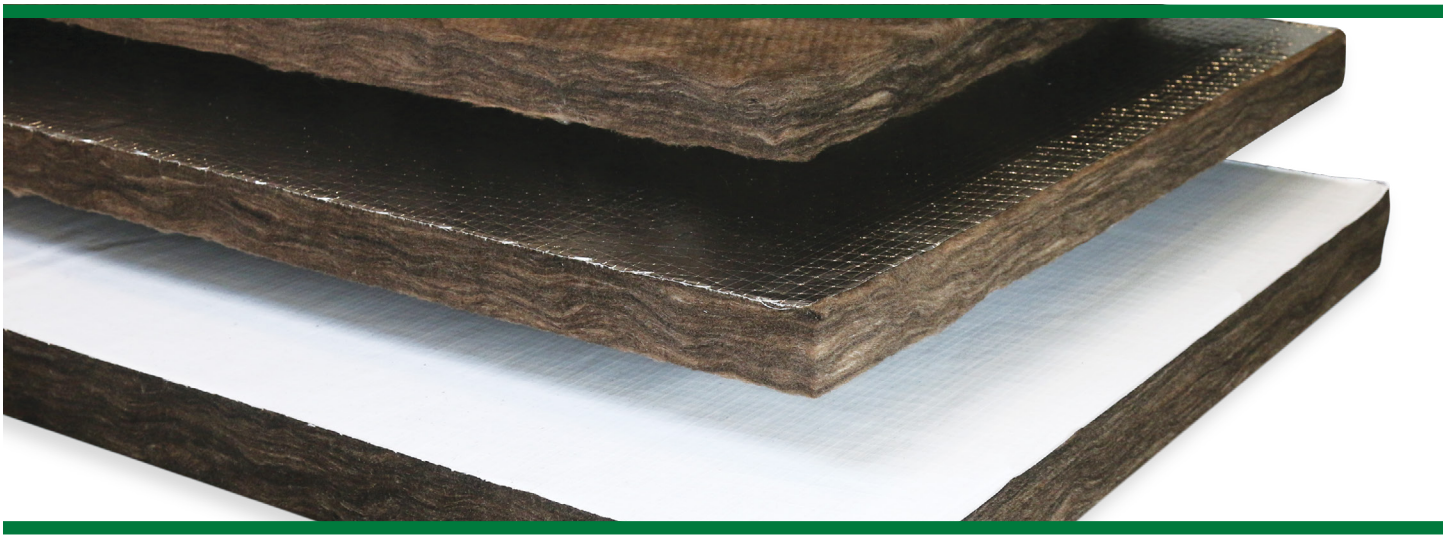
Outdoor Ductwork external insulation

MANSON INSULATION

CONTRACTOR Northern Wolves Inc

JOB NAME Burlington

DATE



AK BOARD™

Temperature Limit: 450° F (232° C)

DESCRIPTION

AK Board is a thermal and acoustical insulation product made from inorganic glass fibers preformed into boards bonded by a thermosetting resin. It is available plain or with factory applied FSK or ASJ facings.

SUSTAINABILITY

Manson Insulation products with ECOSE® Technology are made using our patented, bio-based binder - a smarter alternative to the phenol/formaldehyde (PF) binder traditionally used in fiberglass products. The bio-based binder holds our product together and gives the product its unique appearance.

All of our products are formaldehyde-free and made from sustainable resources, such as recycled glass and sand. And we're proud to be putting glass bottles back to work rather than into landfills. Our products are made with a minimum of 50% recycled glass—totaling an average of 26 million bottles each month.

APPLICATION

Manson Insulation AK Board is a versatile product for thermal and acoustical applications such as heating and air conditioning ducts, power and process equipment, boiler and stack installations, metal and masonry walls, wall and roof panel systems, curtain wall assemblies and cavity walls.

SPECIFICATION COMPLIANCE

- ASTM C612;
- Type IA (1.6, 2.25, 3.0, 4.25, 6.0 PCF) (26, 36, 48, 68, 96 kg/m³)
- Type IB (3.0, 4.25, 6.0 PCF) (48, 68, 96 kg/m³)
- ASTM C1136 (facings); Type II, IV (FSK), Type I, II (ASJ)
- California Title 24
- CGSM 51-GP-10M

INDOOR AIR QUALITY

- UL Environment
  - GREENGUARD Certified
  - GREENGUARD Gold Certified
  - Validated to be Formaldehyde-Free
- EUCEB certified

FIBERGLASS AND MOLD

Fiberglass insulation will not sustain mold growth. However, mold can grow on almost any material when it becomes wet and contaminated. Carefully inspect any insulation that has been exposed to water. If it shows any sign of mold, it must be discarded. If the material is wet but shows no evidence of mold, it should be dried rapidly and thoroughly. If it shows signs of facing degradation from wetting, it should be replaced.

NOTES

The chemical and physical properties of this product represent average values determined in accordance with accepted test methods. The data is subject to normal manufacturing variations. The data is supplied as a technical service and is subject to change without notice. References to numerical flame spread ratings are not intended to reflect hazards presented by these or any other materials under actual fire conditions.

Check with your Manson Insulation Area Manager to ensure information is current.

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TECHNICAL DATA

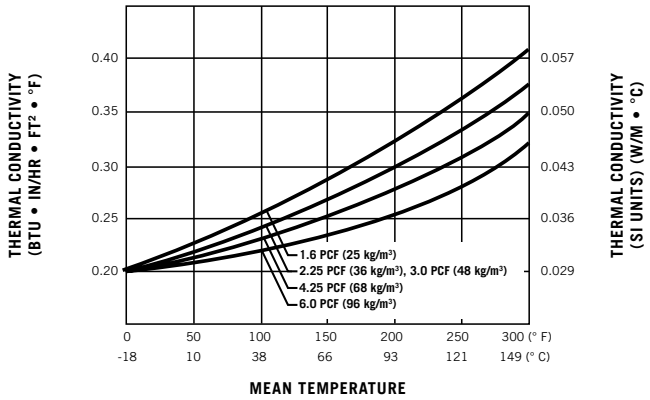
PROPERTY (UNIT)	TEST	PERFORMANCE
Corrosiveness	ASTM C665	Does not accelerate corrosion of steel
Corrosion	ASTM C1617	Pass
Linear Shrinkage	ASTM C356	Less than 0.3%
Maximum Service Temperature	C411	450° F (232° C)
Water Vapor Sorption (by weight)	ASTM C1104	Less than 5%
Water Vapor Permeance	ASTM E96, Procedure A	0.02 perms
Puncture Resistance	TAPPI T803, Beach Units	FSK facing: 25, ASJ facing: 50
Microbial Growth	ASTM C1338	Pass
Surface Burning Characteristics (flame spread/smoke developed)	ASTM E84, UL 723, CAN/ULC S102, NFPA 90A and 90B	UL/ULC Classified FHC 25/50

FORMS AVAILABLE*

DENSITY	THICKNESS	R-VALUE	(R-SI)
1.6 PCF (26 kg/m³)	1½" (38 mm)	R-6.3	(1.1)
	2" (51 mm)	R-8.3	(1.5)
	3" (76 mm)	R-12.5	(2.2)
2.25 PCF (36 kg/m³)	1" (25 mm)	R-4.3	(0.8)
	1½" (38 mm)	R-6.5	(1.1)
	2" (51 mm)	R-8.7	(1.5)
	3" 76 mm)	R-13	(2.3)
	4" (102 mm)	R-17.4	(3.1)
3.0 PCF (48 kg/m³)	1" (25 mm)	R-4.3	(0.8)
	1½" (38 mm)	R-6.5	(1.1)
	2" (51 mm)	R-8.7	(1.5)
	2½" (64 mm)	R-10.9	(1.9)
	3" (76 mm)	R-13	(2.3)
	4" (102 mm)	R-17.4	(3.1)
4.25 PCF † (68 kg/m³)	1" (25 mm)	R-4.3	(0.8)
	1½" (38 mm)	R-6.5	(1.1)
	2" (51 mm)	R-8.7	(1.5)
	2½" (64 mm)	R-10.9	(1.9)
6.0 PCF † (96 kg/m³)	1" (76 mm)	R-4.5	(0.8)
	1½" (89 mm)	R-6.8	(1.2)
	2" (102 mm)	R-9.1	(1.6)

*Available in widths of 24" (610 mm) and 48" (1,219 mm) and lengths from 36" to 120" (915 mm to 3,048 mm).
†Cartons only.

THERMAL EFFICIENCY | ASTM C177



MEAN TEMPERATURE	1.6 PCF		3.0 PCF		6.0 PCF	
	K	K(SI)	K	K(SI)	K	K(SI)
75° F (24° C)	0.24	0.035	0.23	0.033	0.22	0.032
100° F (38° C)	0.25	0.036	0.24	0.035	0.23	0.033
200° F (93° C)	0.33	0.048	0.29	0.042	0.27	0.039
300° F (149° C)	0.42	0.061	0.37	0.053	0.34	0.049

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SOUND ABSORPTION COEFFICIENTS | ASTM C423, TYPE A MOUNTING

			⅓ OCTAVE BAND CENTER FREQUENCY (CYCLES/SEC.)						
DENSITY	FACING	THICKNESS	125	250	500	1000	2000	4000	NRC
1.6 PCF (26 kg/m³)	Plain	1½" (38 mm)	0.19	0.44	0.86	0.98	1.00	1.02	0.80
		2" (51 mm)	0.31	0.57	0.96	1.04	1.03	1.03	0.90
		2½" (64 mm)	0.43	0.82	1.12	1.07	1.04	1.03	1.00
		3" (76 mm)	0.47	0.92	1.17	1.06	1.06	1.04	1.05
2.25 PCF (36 kg/m³)	Plain	1" (25 mm)	0.05	0.24	0.59	0.86	0.97	1.00	0.65
		1½" (38 mm)	0.17	0.49	0.93	1.03	1.03	0.99	0.85
		2" (51 mm)	0.26	0.62	1.05	1.07	1.04	1.05	0.95
	FSK	1" (25 mm)	0.14	0.69	0.81	0.99	0.55	0.27	0.75
		2" (51 mm)	0.63	0.76	1.11	0.75	0.42	0.22	0.75
3.0 PCF (48 kg/m³)	Plain	1" (25 mm)	0.08	0.23	0.62	0.88	0.96	0.99	0.65
		1½" (38 mm)	0.09	0.39	0.89	1.03	1.06	1.01	0.85
		2" (51 mm)	0.29	0.65	1.11	1.13	1.06	1.03	1.00
		3" (76 mm)	0.54	1.01	1.18	1.07	1.07	1.04	1.10
		4" (102 mm)	0.95	1.11	1.17	1.07	1.07	1.06	1.10
	FSK	1" (25 mm)	0.21	0.63	0.84	0.93	0.51	0.22	0.75
		1½" (38 mm)	0.45	0.60	0.99	0.73	0.53	0.27	0.70
		2" (51 mm)	0.67	0.77	0.93	0.74	0.47	0.28	0.75
	ASJ	1" (25 mm)	0.15	0.71	0.65	0.82	0.41	0.16	0.65
		1½" (38 mm)	0.42	0.55	0.91	0.69	0.40	0.23	0.65
		2" (51 mm)	0.75	0.71	0.80	0.66	0.41	0.24	0.65
4.25 PCF (68 kg/m³)	Plain	1" (25 mm)	0.06	0.24	0.69	0.99	1.05	1.02	0.75
	ASJ	2½" (64 mm)	0.75	0.63	0.63	0.62	0.41	0.25	0.55
6.0 PCF (96 kg/m³)	Plain	1" (25 mm)	0.05	0.26	0.77	1.04	1.04	1.03	0.80
		1½" (38 mm)	0.61	0.47	0.78	0.61	0.51	0.35	0.60
		2" (51 mm)	0.13	0.58	1.01	1.05	1.00	1.01	0.90
	FSK	1" (25 mm)	0.23	0.65	0.39	0.48	0.47	0.32	0.50
		1½" (38 mm)	0.61	0.47	0.78	0.61	0.51	0.35	0.60
		2" (51 mm)	0.77	0.50	0.72	0.58	0.53	0.41	0.60
	ASJ	1½" (38 mm)	0.60	0.46	0.62	0.48	0.47	0.31	0.50
		2" (51 mm)	0.77	0.44	0.60	0.50	0.41	0.30	0.50

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Manufactured by Knauf Insulation
09-21

Science.
Applied to Life.™

Roof Ductwork Weater-Proof Jacketing

3M™ VentureClad™ Insulation Jacketing System

Let's redefine the way we build.

Let's Build Smarter.

3M™ VentureClad™ Insulation Jacketing System

Your most complex projects, simplified.

Productivity
Cost effective & efficient

Strength
Trusted durability

Performance
Zero permeability

Insulating a building's pipes and ducts is key to limiting undesired heat gain or loss, resulting in increased energy consumption. For insulation to remain effective, it is critical that it not be compromised by moisture intrusion. This can lead to decreased energy efficiency, performance issues, higher maintenance costs and corrosion under insulation.

So, we're redefining the way we build with a game-changing insulation jacketing system. 3M™ VentureClad™ Insulation Jacketing System helps you save time and labor costs with quick and easy installation and year-round application. Simplify your insulation process by using a zero permeability jacketing to help prevent moisture intrusion and corrosion. Experience the advantages of a durable, long-lasting solution for strength you can trust. We'll help you meet your most complex projects — and your biggest challenges — head-on.

Together, we can build smarter.

Redefine productivity.

- Quick and simple install with no specialized tools, no sharp edges, fasteners or banding
- Can be applied in temps as low as -10°F with service range of -40°F to 248°F
- Easily cut with standard scissors or utility knife
- Simple installation for reduced labor costs
- No glues, primers, activators or heat guns

Redefine strength.

- Highly resistant to punctures and tears
- Backed by a 10-year 3M warranty
- Durable, multi-ply laminate design
- Adhesion in cold, hot and humid conditions
- Easy to patch and repair

Redefine performance.

- Zero permeability, creating a moisture and vapor barrier
- Helps prevent corrosion and mold under insulation
- Excellent chemical resistance for protection in process areas
- Available in multiple colors, textures and sizes
- Proven, certified and compliant
  - UL 723 classified/ASTM E 84
  - Greenguard Gold Certified — Qualifies for LEED credits

A variety of solutions built for your needs.

For Most Standard Applications

3M™ VentureClad™ Insulation Jacketing System
1577CW Series

	White Smooth	1577CW-WM
	White Embossed	1577CW-WME
	Natural Aluminum Smooth	1577CW
	Natural Aluminum Embossed	1577CW-E

Excellent Chemical Resistance: 1577CW-WM/1577CW-WME

Emittance Requirements: 1577CW-WM or 1577CW-WME (White; Med.-High)/1577CW or 1577CW-E (Natural Aluminum; Low)

For Heavy-Duty Applications

3M™ VentureClad™ Heavy Duty Insulation Jacketing System
1579GCW Series

	Heavy-Duty White Smooth	1579GCW-WM
	Heavy-Duty White Embossed	1579GCW-WME
	Heavy-Duty Natural Aluminum Smooth	1579GCW
	Heavy-Duty Natural Aluminum Embossed	1579GCW-E

Product technical data.

All products are classified as zero permeability.

Products	Product Thickness (excluding liner) mils (mm)	Peel Adhesion oz/in (N/25mm)	Tensile lb/in (N/25mm)	Puncture lb (N)
3M™ VentureClad™ Insulation Jacketing System – 1577 Series				
1577CW-WM - White, Smooth	7 (0.18)	80 (22)	108 (480)	33 (147)
1577CW-WME - White, Embossed	9 (0.23)	82 (23)	100 (445)	44 (194)
1577CW Natural Aluminum, Smooth	6 (0.16)	75 (21)	80 (355)	25 (112)
1577CW-E - Natural Aluminum, Embossed	9 (0.23)	83 (23)	74 (329)	27 (121)
3M™ VentureClad™ Insulation Jacketing Tape System – 1579 Series				
1579GCW-WM - White, Smooth	15 (0.38)	102 (28)	184 (818)	57 (253)
1579GCW-WME - White, Embossed	16 (0.40)	126 (35)	184 (818)	57 (253)
1579GCW - Natural Aluminum, Smooth	15 (0.38)	87 (24)	155 (690)	47 (210)
1579GCW-E - Natural Aluminum, Embossed	16 (0.40)	73 (20)	149 (663)	49 (218)

Learn how 3M's advanced technologies can help you build smarter. Visit 3M.com/VentureClad or contact your 3M representative.

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